

Atomic Physics

Single Correct Type Questions

1. The angular momentum for the electron in Bohr's orbit is L . If the electron is assumed to revolve in second orbit of hydrogen atom, then the change in angular momentum will be:
[10 April, 2023 Shift-I]

(a) $\frac{L}{2}$ (b) zero
(c) L (d) $2L$

2. An electron of a hydrogen like atom, having $Z = 4$, jumps from 4th energy state to 2nd energy state, The energy released in this process, will be:
(Given $Rch = 13.6\text{ eV}$)

Where R = Rydberg constant
 c = Speed of light in vacuum

h = Planck's constant [1 Feb, 2023 (Shift-II)]

(a) 13.6 eV (b) 10.5 eV
(c) 3.4 eV (d) 40.8 eV

3. The radius of electron's second stationary orbit in Bohr's atom is R . The radius of 3rd orbit will be

[31 Jan, 2023 (Shift-II)]

(a) $R/3$ (b) $2.25R$
(c) $3R$ (d) $9R$

4. A small particle of mass m moves in such a way that its potential energy $U = \frac{1}{2}m\omega^2 r^2$ where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of momentum and circular orbit, the radius of n^{th} orbit will be proportional to.

[6 April, 2023 (Shift-II)]

(a) $\sqrt{n}$ (b) n
(c) n^2 (d) $\frac{1}{n}$

5. The ratio for the speed of the electron in the 3rd orbit of He^+ to the speed of the electron in the 3rd orbit of hydrogen atom will be:
[25 June, 2022 (Shift-I)]

(a) 1:1 (b) 1:2
(c) 4:1 (d) 2:1

6. The momentum of an electron revolving in n^{th} orbit is given by: (Symbols have their usual meanings)
[25 July, 2022 (Shift-I)]

(a) $\frac{nh}{2\pi r}$ (b) $\frac{nh}{2r}$
(c) $\frac{nh}{2\pi}$ (d) $\frac{2\pi r}{nh}$

7. Imagine that the electron in a hydrogen atom is replaced by a muon (μ). The mass of muon particle is 207 times that of an electron and charge is equal to the charge of an electron. The ionization potential of this hydrogen atom will be
[18 March, 2021 (Shift-I)]

(a) 13.6 eV (b) 2815.2 eV
(c) 331.2 eV (d) 27.2 eV

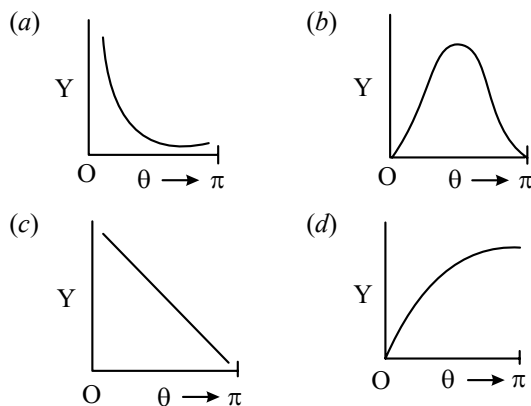
8. The graph which depicts the results of Rutherford gold foil experiment with α -particles is

θ : Scattering angle

Y: Number of scattered α -particles detected

(Plots are schematic and not to scale)

[8 Jan, 2020 (Shift-I)]



9. A hydrogen atom, initially in the ground state is excited by absorbing a photon of wavelength $980 \text{ \AA}$. The radius of the atom in the excited state, in terms of Bohr radius a_0 , will ($hc = 12500 \text{ eV} - \text{\AA}$) [11 Jan, 2019 (Shift-I)]
- (a) $25 a_0$ (b) $9 a_0$
(c) $16 a_0$ (d) $4 a_0$
10. A He^+ ion is in its first excited state. Its ionization energy is: [9 April, 2019 (Shift-II)]
- (a) 6.04 eV (b) 13.60 eV
(c) 54.40 eV (d) 48.36 eV
11. A photon is emitted in transition from $n = 4$ to $n = 1$ level in hydrogen atom. The corresponding wavelength for this transition is (given, $h = 4 \times 10^{-15} \text{ eVs}$): [24 Jan, 2023 (Shift-II)]
- (a) 94.1 nm (b) 941 nm
(c) 97.4 nm (d) 99.3 nm
12. Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level. [29 July, 2022 (Shift-I)]
- (a) $3 : 4$
(b) $4 : 3$
(c) $1 : 4$
(d) $4 : 1$
13. Hydrogen atom from excited state comes to the ground state by emitting a photon of wavelength λ . The value of principal quantum number ' n ' of the excited state will be: (R: Rydberg constant) [25 July, 2022 (Shift-II)]
- (a) $\sqrt{\frac{\lambda R}{\lambda - 1}}$ (b) $\sqrt{\frac{\lambda R}{\lambda R - 1}}$
(c) $\sqrt{\frac{\lambda}{\lambda R - 1}}$ (d) $\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$
14. According to Bohr atom model, in which of the following transitions will the frequency be maximum? [24 Feb, 2021 (Shift-II)]
- (a) $n = 3$ to $n = 2$ (b) $n = 4$ to $n = 3$
(c) $n = 2$ to $n = 1$ (d) $n = 5$ to $n = 4$
15. If λ_1 and λ_2 are the wavelengths of the third member of Lyman and first member of the Paschen series respectively, then the value of $\lambda_1 : \lambda_2$ is: [26 Feb, 2021 (Shift-I)]
- (a) $1 : 3$ (b) $7 : 108$
(c) $1 : 9$ (d) $7 : 135$
16. The energy required to ionise a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state: [9 Jan, 2020 (Shift-II)]
- (a) 11.4 nm (b) 24.2 nm
(c) 35.8 nm (d) 8.6 nm
17. In Li^{++} , electron in first Bohr orbit is excited to a level by a radiation of wavelength λ . When the ion gets deexcited to the ground state in all possible ways (including intermediate emission) a total of six spectral lines are observed. What is the value of λ ? (Given: $h = 6.63 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ ms}^{-1}$) [10 April, 2019 (Shift-II)]
- (a) 10.8 nm (b) 11.4 nm
(c) 9.4 nm (d) 12.3 nm
18. The electron in a hydrogen atom first jumps from the third excited state to the second excited state and subsequently to the first excited state. The ratio of the respective wavelengths, λ_1/λ_2 , of the photons emitted in this process is: [12 April, 2019 (Shift-II)]
- (a) $20/7$ (b) $7/5$
(c) $9/7$ (d) $27/5$
19. An excited He^+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm , in making a transition to ground state. The quantum number n , corresponding to its initial excited state is (for photon of wavelength λ , energy $E = \frac{1240 \text{ eV}}{\lambda(\text{in nm})}$): [12 April, 2019 (Shift-I)]
- (a) $n = 4$ (b) $n = 6$
(c) $n = 5$ (d) $n = 7$
20. The waves emitted when a metal target is bombarded with high energy electrons are [8 April, 2023 (Shift-II)]
- (a) Microwaves (b) X-rays
(c) Infrared rays (d) Radio Waves

Integer Type Questions

21. A light of energy 12.75 eV is incident on a hydrogen atom in its ground state. The atom absorbs the radiation and reaches to one of its excited states. The angular momentum of the atom in the excited state is $\frac{x}{\pi} \times 10^{-17} \text{ eVs}$. The value of x is _____ (use $h = 4.14 \times 10^{-15} \text{ eVs}$, $c = 3 \times 10^8 \text{ ms}^{-1}$).

[1 Feb, 2023 (Shift-I)]

22. The radius of 2nd orbit of He⁺ of Bohr's model is r_1 and that of fourth orbit of Be³⁺ is represented as r_2 . Now the ratio r_2/r_1 is $x : 1$. The value of x is _____.

[13 April, 2023 (Shift-I)]

23. $\sqrt{d_1}$ and $\sqrt{d_2}$ are the impact parameters corresponding to scattering angles 60° and 90° respectively, when an α particle is approaching a gold nucleus. For $d_1 = x d_2$, the value of x will be _____. [29 June, 2023 (Shift-I)]

24. The first member of the Balmer series of hydrogen atom has a wavelength of 6561 Å. The wavelength of the second member of the Balmer series (in nm) is _____.



ANSWER KEY

1. (c) 2. (d) 3. (b) 4. (a) 5. (d) 6. (a) 7. (b) 8. (a) 9. (c) 10. (b)
 11. (a) 12. (a) 13. (b) 14. (c) 15. (d) 16. (a) 17. (a) 18. (a) 19. (c) 20. (b)
 21. [828] 22. [2] 23. [3] 24. [486]

EXPLANATIONS

1. (c) Angular momentum, $L = \frac{nh}{2\pi}$
 For Bohr orbit is, $L_1 = L = \frac{1 \cdot h}{2\pi}$
 In second orbit of hydrogen atom ($n = 2$),
 $L_2 = \frac{2h}{2\pi} = 2L$
 So, change momentum $= L_2 - L_1 = 2L - L = L$
2. (d) $\because E_n = -\frac{13.6Z^2}{n^2} eV$
 $\Rightarrow \Delta E = 13.6Z^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right] eV$
 $= 13.6 \times (4)^2 \left(\frac{1}{4} - \frac{1}{16} \right) eV$
 $\Rightarrow \Delta E = 40.8 eV$
3. (b) We know that, Radius of n^{th} orbit, $R = 0.529 \frac{n^2}{z} \text{\AA}$
 $\frac{R_3}{R_2} = \frac{9}{4}$
 $R_3 = \frac{9}{4} R_2$
 $\therefore R_3 = 2.25 R$
4. (a) $U = \frac{1}{2} m \omega^2 r^2$
 $F = -\frac{du}{dr} = -m\omega^2 r = \frac{mv^2}{r} \Rightarrow v = \omega r$
 $\therefore mvr = \frac{nh}{2\pi}$
 $\therefore m\omega r^2 = \frac{nh}{2\pi} \Rightarrow r = \sqrt{\frac{nh}{2\pi m\omega}} \Rightarrow r \propto \sqrt{n}$
5. (d) As we know the atomic number He^+ , $Z_{He^+} = 2$
 and the atomic number Hydrogen H^+ , $Z_{H^+} = 1$
 Since, $v \propto \frac{Z}{n}$ ($n = \text{constant}$)

We have;

$$v_{He^+} \propto Z_{He^+} \quad \dots(i)$$

$$\text{and, } v_{H^+} \propto Z_{H^+} \quad \dots(ii)$$

Now, On dividing equation (i) and equation (ii) we have;

$$\frac{v_{He^+}}{v_{H^+}} = \frac{Z_{He^+}}{Z_{H^+}} \Rightarrow \frac{v_{He^+}}{v_{H^+}} = \frac{2}{1}$$

$$6. (a) mvr = \frac{nh}{2\pi}$$

$$\text{Therefore, momentum, } p = mv = \frac{nh}{2\pi r}$$

$$7. (b) E \propto \frac{1}{r}, r \propto \frac{1}{m}$$

$$E \propto m$$

$$\text{Ionization potential} = 13.6 \times \frac{(m_\mu) eV}{(m_e)}$$

$$= 13.6 \times 207 eV = 2815.2 eV$$

8. (a) Results of Rutherford gold foil experiment gives

$$N \propto \frac{1}{\sin^4 \left(\frac{\theta}{2} \right)}$$

9. (c) Energy supplied

$$E = \frac{12500}{980} = 12.75 eV$$

$$\therefore E_n - E_1 = 12.75$$

$$\Rightarrow (13.6) \left(1 - \frac{1}{n^2} \right) = 12.75$$

$$\Rightarrow n^2 \approx 14.3 \Rightarrow n \approx 4$$

$$\text{New radius} = 16 a_0$$

10. (b) Energy levels in Hydrogen like atom is given by

$$E = -13.6 \frac{Z^2}{n^2} eV$$

As He^+ is 1st excited state

$$\therefore z = 2, n = 2$$

$$E = -13.6 \text{ eV}$$

As total energy of He^+ in 1st excited state is -13.6 eV , ionization energy should be $+13.6 \text{ eV}$.

11. (a) For a hydrogen atom, $E = \frac{-13.6 \text{ eV}}{n^2}$

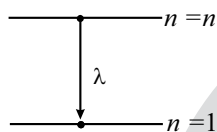
$$\Rightarrow \frac{hc}{\lambda} = \left[1 - \frac{1}{16}\right] (13.6 \text{ eV})$$

$$\Rightarrow \lambda = 94.1 \text{ nm}$$

12. (a) $\Delta E_1 = -\frac{E_0}{4} + \frac{E_0}{1} = \frac{3}{4} E_0$
 $\Delta E_2 = 0 - (-E_0) = E_0$

$$\frac{\Delta E_1}{\Delta E_2} = \frac{3}{4}$$

13. (b) According to definition of wave number, we can write



$$\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$$

$$\Rightarrow 1 - \frac{1}{n^2} = \frac{1}{\lambda R}$$

$$\Rightarrow \frac{1}{n^2} = 1 - \frac{1}{\lambda R} = \frac{\lambda R - 1}{\lambda R} \Rightarrow n = \sqrt{\frac{\lambda R}{\lambda R - 1}}$$

14. (c) We know that, $\Delta E \propto \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$

If $n_1 = 3$ and $n_2 = 2$

$$\left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{9-4}{36} = \frac{5}{36}$$

If $n_1 = 4$ and $n_2 = 3$

$$\left(\frac{1}{3^2} - \frac{1}{4^2} \right) = \frac{16-9}{144} = \frac{7}{144}$$

If $n_1 = 2$ and $n_2 = 1$

$$\left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{4-1}{4} = \frac{3}{4}$$

If $n_1 = 5$ and $n_2 = 4$

$$\left(\frac{1}{4^2} - \frac{1}{5^2} \right) = \frac{25-16}{400} = \frac{9}{400}$$

Since, $\Delta E = h\nu$.

Hence option (c) is the correct answer.

15. (d) The wavelength of spectral line of the third member of Lyman series;

$$\frac{1}{\lambda_1} = RZ^2 \left(\frac{1}{1^2} - \frac{1}{4^2} \right)$$

The wavelength of spectral line of the first member of paschen series;

$$\frac{1}{\lambda_2} = RZ^2 \left(\frac{1}{3^2} - \frac{1}{4^2} \right)$$

$$\frac{1/\lambda_1}{1/\lambda_2} = \frac{1 - \frac{1}{16}}{\frac{1}{9} - \frac{1}{16}} \Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{\frac{15}{16}}{\frac{7}{9 \times 16}} = \frac{15 \times 9}{7} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{7}{135}$$

16. (a) $\frac{hc}{\lambda} = (13.6 \text{ eV}) z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

$$\frac{hc}{\lambda} = (13.6 \text{ eV})(3^2) \left[\frac{1}{1^2} - \frac{1}{3^2} \right]$$

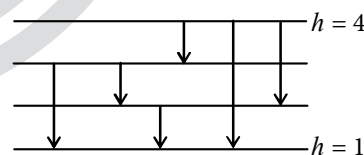
$$\Rightarrow \frac{hc}{\lambda} = (13.6 \text{ eV})(9) \left(\frac{8}{9} \right)$$

$$\text{Wavelength} = \frac{1240}{8 \times 13.6} \text{ nm} = 11.39 \text{ nm}$$

17. (a) $\frac{hc}{\lambda} = 13.6 \text{ eV} (9) \left\{ 1 - \frac{1}{16} \right\}$

$$\frac{1240 \text{ eV}}{\lambda} = \frac{15}{16} \times 9 \times 13.6 \text{ eV}$$

$$\Rightarrow \lambda = 10.8 \text{ nm}$$



Hence, option (a) is correct answer.

18. (a) We know that, $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$;

$$\frac{1}{\lambda_1} = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \Rightarrow \frac{1}{\lambda_1} = R \left(\frac{7}{9 \times 16} \right);$$

$$\frac{1}{\lambda_2} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \Rightarrow \frac{1}{\lambda_2} = R \left(\frac{5}{4 \times 9} \right)$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\frac{5}{36}}{\frac{7}{9 \times 16}} = \frac{5 \times 9 \times 16}{7 \times 36} = \frac{20}{7}$$

19. (c) $E = E_1 + E_2$

$$13.6(2)^2 \left(\frac{1}{1^2} - \frac{1}{n^2} \right) = \frac{1240}{108.5} + \frac{1240}{30.4}$$

$$\left(1 - \frac{1}{n^2} \right) = \frac{52.22}{54.4}$$

$$\frac{1}{n^2} = \frac{2.18}{54.4} \Rightarrow n^2 = 25 \Rightarrow n = 5$$

20. (b) When a target metal is bombarded with high energy electron the X rays are emitted.

21. [828] Ground state energy = -13.6 eV

$$\therefore \frac{-13.6\text{eV}}{n^2} = -13.6 + 12.75$$

$$\Rightarrow \frac{-13.6\text{eV}}{n^2} = -0.85$$

$$\Rightarrow \boxed{n = 4}$$

$$\text{Angular momentum} = \frac{nh}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi}$$

$$\Rightarrow \text{Angular momentum} = \frac{2}{\pi} \times 4.14 \times 10^{-15} \\ = \frac{828 \times 10^{-17}}{\pi} \text{ eVs}$$

22. [2] $r \propto \frac{n^2}{z} \Rightarrow \frac{r_{He^+}}{r_{Be^{3+}}} = \left(\frac{\eta_{He^+}}{\eta_{Be^{3+}}} \right)^2 \times \frac{z_{Be^{3+}}}{z_{He^+}}$

$$\frac{r_{He^+}}{r_{Be^{3+}}} = \left(\frac{2}{4} \right)^2 \times \left(\frac{4}{2} \right) = \frac{1}{2}$$

23. [3] We know that, $d \propto \cot^2 \frac{\theta}{2} \Rightarrow \sqrt{d} \propto \cot \frac{\theta}{2}$

$$\frac{\sqrt{d_1}}{\sqrt{d_2}} = \frac{\cot \frac{\theta_1}{2}}{\cot \frac{\theta_2}{2}} \Rightarrow \sqrt{x} = \frac{\cot 30^\circ}{\cot 45^\circ} \Rightarrow \sqrt{x} = \sqrt{3} \Rightarrow x = 3$$

24. [486] We have, $\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

For first member of Balmer series, $n_1 = 2, n_2 = 3$

$$\frac{1}{\lambda_1} = R(1)^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36}$$

For second member of Balmer series, $n_1 = 2, n_2 = 4$

$$\frac{1}{\lambda_2} = R(1)^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3R}{16}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{20}{27}$$

$$\lambda_2 = \frac{20}{27} \times 6561 \text{ \AA} = 4860 \text{ \AA}$$

Hence, wavelength of the second member of the Balmer series is 486 nm.